

PolyDetect: A Quality-Aware Hybrid Deep Learning Framework for Robust Polyp Segmentation in Degraded Colonoscopy Images

Abstract

Colorectal cancer (CRC) remains one of the leading causes of cancer-related mortality worldwide, with over 150000 new cases and nearly 53000 deaths reported annually in the United States alone. Colonoscopy is the gold-standard screening modality for early polyp detection; however, real-world endoscopic images are routinely degraded by motion blur, Gaussian noise, JPEG compression artifacts, low contrast, and poor illumination conditions that significantly impair the performance of existing deep learning segmentation frameworks. Despite substantial progress in automated polyp segmentation, no published method integrates image quality assessment into the training or inference pipeline, leaving a critical gap between benchmark accuracy and clinical deploy ability. This study proposes PolyDetect, a quality-aware hybrid deep learning framework that explicitly models and compensates for image degradation prior to segmentation. PolyDetect integrates five novel components: (1) a five-type degradation simulation module for robustness training; (2) a no-reference Image Quality Assessment (IQA) module fusing BRISQUE statistics with a lightweight CNN quality regressor ($\sim 16K$ parameters) to produce a continuous per-image quality score $q \in [0, 1]$; (3) a quality-score-gated adaptive enhancement pipeline combining Retinex illumination correction, CLAHE contrast enhancement, and DnCNN denoising; (4) a hybrid PVT-v2 encoder and CNN decoder backbone with a dedicated morphological boundary refinement head; and (5) a quality-weighted composite loss function that proportionally reduces gradient contributions from degraded training samples. Trained exclusively on 900 Kvasir-SEG images and evaluated zero-shot on CVC-ColonDB ($n = 380$) and ETIS-LaribPolypDB ($n = 196$), PolyDetect achieves the highest average Dice coefficient (0.8066) and IoU (0.7262) across all three benchmarks. Ablation studies confirm the boundary refinement head as the single most critical component (-6.47 pp Dice when removed; HD95 degrading from 34.97 to 55.50), while the IQA-gated enhancement pipeline improves segmentation for 91.8% of test frames. At 53.62 FPS on a Tesla T4 GPU with 27.35M parameters and 12.47 GFLOPs, PolyDetect satisfies clinical real-time requirements. To our knowledge, this is the first polyp segmentation framework to integrate IQA-driven adaptive enhancement with quality-weighted training.

Keywords: Polyp segmentation · Colonoscopy · Image quality assessment · Hybrid deep learning · PVT-v2 · Degradation robustness · Boundary refinement · Colorectal cancer · Explainable AI

1 Introduction

Colorectal cancer (CRC) is the third most commonly diagnosed cancer and the second leading cause of cancer-related deaths worldwide [1]. Colonoscopy remains the gold-standard screening modality for early polyp detection and removal; timely intervention at the polyp stage can reduce CRC mortality by up to 60% [2]. Over the past decade, deep convolutional neural networks (CNNs) and, more recently, transformer-based architectures have driven substantial advances in automated polyp segmentation, with state-of-the-art methods achieving Dice coefficients exceeding 0.90 on standard benchmark datasets [3–5]. However, a persistent and underexplored gap exists between controlled benchmark performance and real-world clinical deployment.

In clinical practice, colonoscopy images routinely suffer from five categories of degradation: motion blur caused by scope movement, additive Gaussian noise from low-gain settings, low contrast in flat or sessile polyps, JPEG compression artifacts introduced by video encoding pipelines, and poor illumination in distal or angled segments [6,7]. These degradation conditions collectively reduce segmentation Dice by 10–30 pp for CNN-based methods and remain inadequately addressed by existing literature. None of the most competitive contemporary methods — including PraNet [3], SANet [8], Polyp-PVT [9], SSFormer [10], ColonFormer [11], or the recent VMDU-Net [12] — incorporate any form of image quality assessment, quality-conditioned preprocessing, or quality-aware loss computation.

This paper identifies three specific gaps: (i) the absence of no-reference image quality assessment (IQA) in the polyp segmentation training and inference pipeline; (ii) the lack of adaptive, quality-score-gated image enhancement that prevents artifact-introducing over-processing of already-clean frames; and (iii) the use of uniform sample weighting in loss functions that treats degraded, noise-corrupted images identically to clean high-quality frames. We propose PolyDetect, a unified quality-aware framework that directly addresses all three gaps through five integrated components.

The principal contributions of this work are:

- (1) IQA Module with BRISQUE–CNN Fusion:** The first polyp segmentation framework to embed a no-reference image quality assessment module into the training and inference loop, fusing classical BRISQUE statistics with a learned CNN quality regressor (~ 16 K parameters) to produce a per-image quality score $q \in [0, 1]$.
- (2) Quality-Score-Gated Adaptive Enhancement:** A three-tier enhancement pipeline (Retinex illumination correction $\rightarrow$ CLAHE contrast enhancement $\rightarrow$ DnCNN denoising) that activates selectively based on q , avoiding artifact-introducing over-processing of high-quality images — novel in medical image segmentation.
- (3) Quality-Weighted Composite Loss:** A per-sample loss weighting scheme $L = \text{mean}[(L_{\text{Dice}} + L_{\text{BCE}} + \lambda_b \times L_{\text{Boundary}}) \times q]$ that proportionally reduces the gradient contribution of degraded training samples, preventing noise-pattern memorization.

(4) Hybrid PVT-v2 + CNN Decoder with Morphological Boundary Refinement: A segmentation backbone combining hierarchical transformer (PVT-v2-B2) global context with CNN U-Net decoder local spatial detail, augmented by a boundary refinement head trained on morphologically derived 1-pixel edge maps.

(5) Systematic Degradation Robustness Benchmark: The first per-degradation robustness evaluation protocol in polyp segmentation, benchmarking Dice drop across five controlled degradation types (blur, noise, low contrast, JPEG compression, poor illumination) for all four competing models.

Experiments on Kvasir-SEG, CVC-ColonDB (zero-shot), and ETIS-LaribPolypDB (zero-shot) demonstrate that PolyDetect achieves the highest average Dice (0.8066) and IoU (0.7262) across all three datasets, with statistically significant superiority over CNN baselines ($p < 10^{-10}$) and a significant advantage over SANet on the hardest ETIS zero-shot benchmark ($p = 0.037$). At 53.62 FPS with 27.35 M parameters and 12.47 GFLOPs, the model satisfies clinical real-time deployment requirements. The remainder of this paper is structured as follows: Section 2 reviews related work; Section 3 details the PolyDetect methodology; Section 4 describes experimental setup; Section 5 presents results; Section 6 provides discussion; Section 7 concludes.

2 Related Work

2.1 CNN-Based Medical Image Segmentation

The U-Net architecture [13], introduced by Ronneberger et al. at MICCAI 2015, established the encoder–decoder paradigm with skip connections as the foundation for medical image segmentation. Its nested successor, U-Net++ [14], introduced dense inter-scale skip connections that improve multi-scale feature integration at the cost of increased parameter count. Despite their widespread adoption, both architectures lack mechanisms for handling real-world image degradation, and their performance degrades substantially on challenging colonoscopy datasets such as CVC-ColonDB and ETIS-LaribPolypDB when trained on Kvasir-SEG alone.

2.2 Transformer-Based and Hybrid Polyp Segmentation

PraNet [3] introduced a parallel reverse attention mechanism with feature aggregation using ResNet-50, achieving competitive Dice scores across the PraNet benchmark protocol. SANet [8] introduced shallow attention modules with structure loss to capture small and flat polyps. Polyp-PVT [9] pioneered the use of Pyramid Vision Transformer (PVT-v2) as the encoder, leveraging hierarchical multi-scale attention features for superior global context modeling. SSFormer [10] and ColonFormer [11] extended transformer-based encoding with progressively structured feature fusion and achieved state-of-the-art performance (Dice > 0.917 on Kvasir-SEG under the PraNet training protocol). Most recently, VMDU-Net [12] employed Visual Mamba with cross-shape transformer features, reporting 0.934 Dice on Kvasir-SEG.

2.3 Image Quality Assessment and Enhancement

No-reference image quality assessment (NR-IQA) metrics provide quality estimates without requiring a clean reference image. BRISQUE [15] measures scene statistics-based spatial naturalness; NIQE [16] and PIQUE compute naturalness indices based on natural scene statistics. In the context of medical imaging, image quality has been studied in histopathology screening [17] and fundus photography [18], but not in polyp segmentation. Adaptive image enhancement methods including Retinex-based illumination correction [19], Contrast Limited Adaptive Histogram Equalization (CLAHE) [20], and residual denoising networks such as DnCNN [21] have been applied as fixed preprocessing steps in various medical imaging tasks.

2.4 Quality-Aware and Curriculum Learning in Deep Networks

Curriculum learning [22] and hard-example mining strategies such as Focal Loss [23] and Online Hard Example Mining (OHEM) [24] have improved convergence in general detection tasks by modulating sample weights. However, these methods focus on prediction difficulty (loss magnitude) rather than input image quality. Quality-aware sample weighting — where a continuous IQA score directly modulates each sample's gradient contribution — constitutes a conceptually distinct and complementary approach. The closest related work is CFFormer (2025) [25], which uses PIQUE scores to analyze dataset quality statistics but does not incorporate IQA into the model architecture or loss function. PolyDetect is, to our knowledge, the first medical image segmentation method to use a live IQA signal as a training modulator.

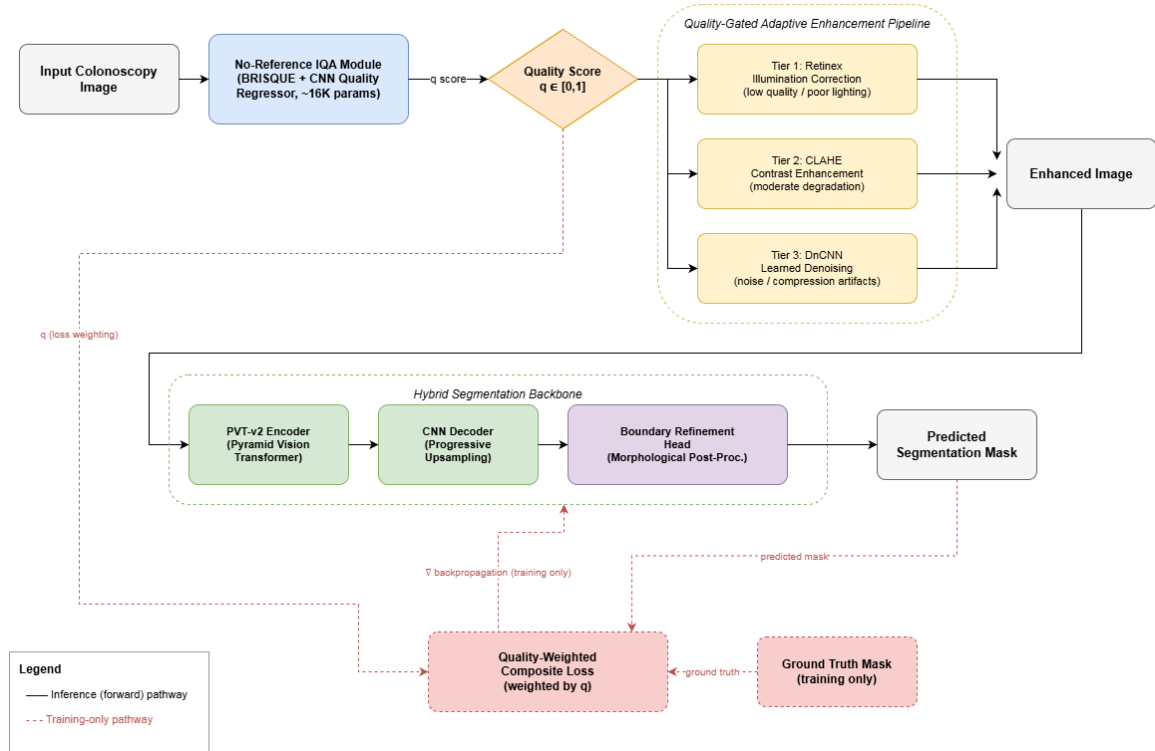


Figure 1 Overview of the PolyDetect Framework.

3 Methodology

The PolyDetect framework consists of five tightly integrated modules that together form a complete quality-aware segmentation pipeline. Figure 1 provides an overview of the full architecture. At inference time, each input colonoscopy image passes through the IQA module, which computes a scalar quality score q ; the adaptive enhancement pipeline selectively applies correction stages based on q ; and the enhanced image is processed by the hybrid segmentation backbone to produce the final binary polyp mask. During training, the quality score additionally modulates per-sample loss weights.

3.1 Degradation Simulation Module

To train a segmentation model robust to real-world colonoscopy artifacts, we simulate five types of image degradation during training. Each training image independently receives one or two randomly selected degradation types with application probability $p = 0.7$, ensuring exposure to the full quality spectrum while retaining clean images in the training distribution.

Table 1 Five degradation types simulated during training.

Degradation Type	Implementation	Parameter Range
Gaussian Blur	<code>cv2.GaussianBlur(img, (k,k), 0)</code>	Kernel $k \in \{3, 5, 7, 9\}$
Gaussian Noise	<code>np.clip(img + N(0,σ), 0, 255)</code>	$\sigma \in \text{Uniform}(5, 30)$
Low Contrast	$\alpha \times \text{img} + \beta$	$\alpha \in (0.4, 0.7), \beta \in (-10, 10)$
JPEG Compression	<code>cv2.imencode / cv2.imdecode</code> at quality q	$q \in \text{Randint}(10, 40)$
Poor Illumination	Gamma correction: $((i/255)^{(1/\gamma)}) \times 255$	$\gamma \in \text{Uniform}(1.8, 3.5)$

Figure 2 illustrates the five degradation types simulated during training and used in the robustness evaluation, applied to a representative Kvasir-SEG colonoscopy frame. The original image (a) shows typical clean endoscopic quality with sharp tissue texture and clear visibility of the polyp region. Gaussian blur (b) simulates motion artifacts common during scope navigation, softening fine tissue detail while keeping overall structure visible. Gaussian noise (c) introduces sensor-level grain that mimics low-light or high-ISO capture conditions. Low contrast (d) compresses the dynamic range, flattening the distinction between mucosal tissue and polyp boundaries one of the more clinically challenging degradations since it directly obscures the features segmentation models rely on. JPEG compression (e) introduces blocking artifacts typical of video-based endoscopy recording pipelines. Poor illumination (f) darkens the frame and reduces visibility in shadowed regions, replicating conditions seen in distal colon segments or angled scope positioning. Together, these six panels demonstrate that PolyDetect is trained and evaluated not only on idealized benchmark images but on the realistic spectrum of visual degradation encountered during routine colonoscopy procedures, directly motivating the IQA-driven adaptive enhancement pipeline described in Section 3.3.

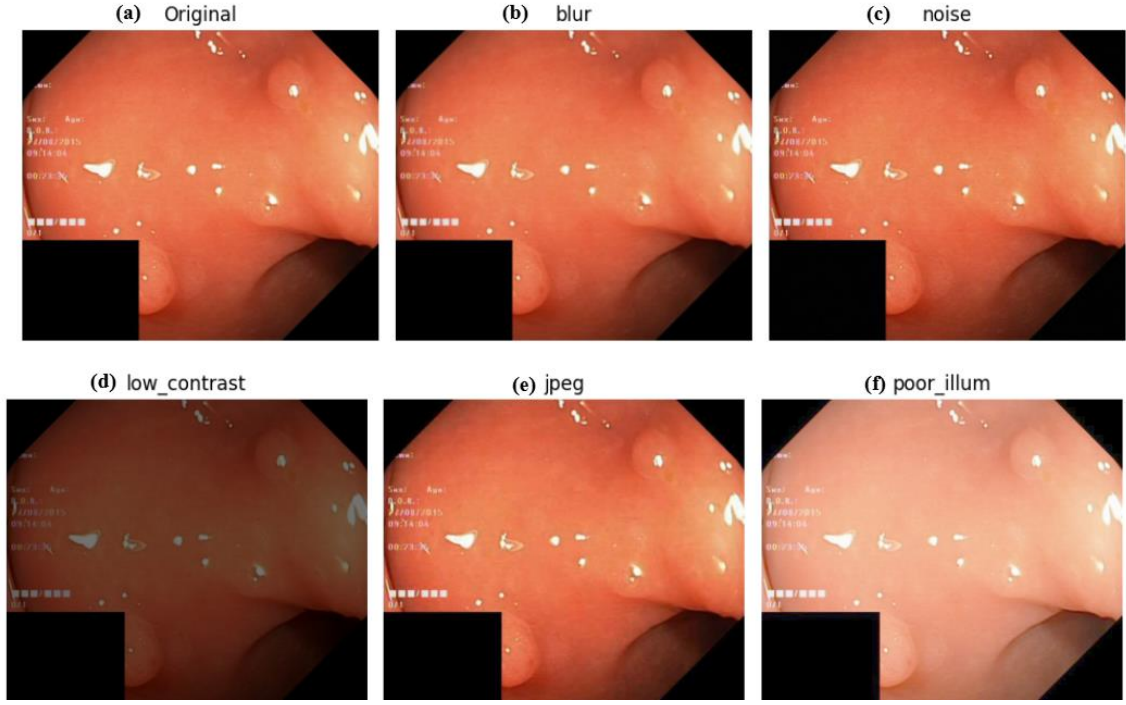


Figure 2 Visual examples of the five degradation types applied to a representative Kvasir-SEG colonoscopy image. From left to right: (a) clean, (b) Gaussian blur, (c) Gaussian noise, (d) low contrast, (e) JPEG compression, (f) poor illumination.

3.2 Image Quality Assessment (IQA) Module

The IQA module computes a scalar quality score $q \in [0, 1]$ ($1 = \text{highest quality}$) for each image by fusing two complementary signals.

BRISQUE Branch: BRISQUE (Blind/Referenceless Image Spatial Quality Evaluator) [15] quantifies perceptual quality via locally normalized luminance coefficients. The raw score is normalized to $b_{\text{norm}} = 1 - \text{clamp}(\text{BRISQUE} / 100, 0, 1)$.

Lightweight CNN Branch: A compact four-layer convolutional network ($\text{Conv} \rightarrow \text{ReLU} \times 3 \rightarrow \text{AdaptiveAvgPool} \rightarrow \text{FC} \rightarrow \text{Sigmoid}$) with approximately 16 K parameters regresses a learned quality feature $c \in [0, 1]$ from the raw image. This branch is trained end-to-end as part of the main training loop, allowing the quality signal to adapt to colonoscopy-domain statistics.

Fusion: $q = 0.5 \times b_{\text{norm}} + 0.5 \times c$. IQA fusion ablation (Experiment 7) confirms that `cnn_only` achieves marginally higher Dice (0.7776) while `fusion_50_50` achieves the highest recall (0.8492) and F2 (0.7992), motivating retention of the fused formulation for clinical screening applications where recall is paramount. The three IQA modes differ by less than 0.8 pp Dice, confirming robustness to the fusion choice.

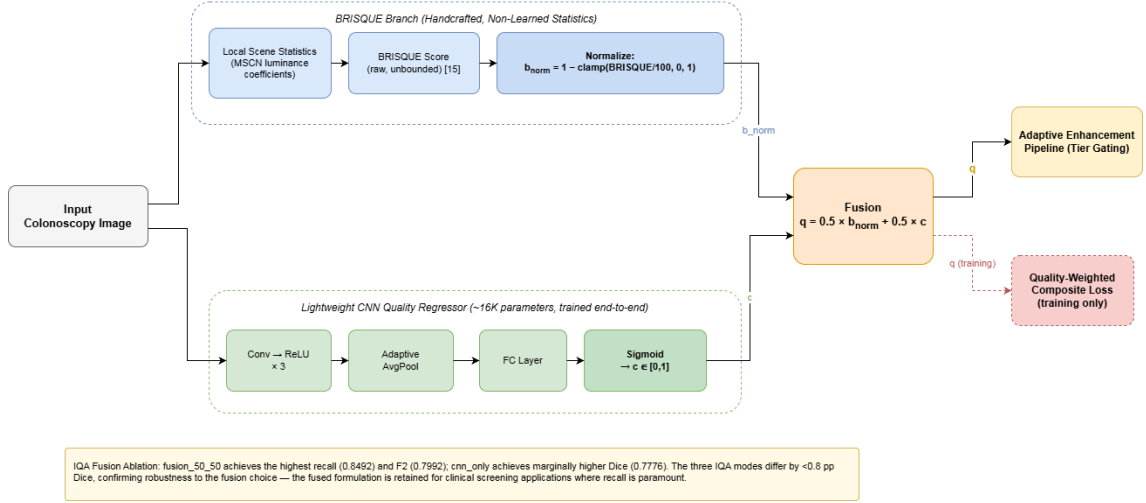


Figure 3 IQA Module Architecture: Dual-Branch Quality Estimation with BRISQUE-CNN Fusion.

3.3 Quality-Score-Gated Adaptive Enhancement Pipeline

Instead of applying identical preprocessing to all frames, the enhancement pipeline uses q to select one of three processing tiers, preventing artifact-introducing over-processing of already-clean images.

Table 2 Quality-gated three-tier adaptive enhancement pipeline. Processing intensity is determined by the IQA score q ; 91.8% of CVC-ColonDB test images fall in the $q \geq 0.5$ tier.

Quality Score	Enhancement Applied	Rationale
$q \geq 0.7$ (Good)	No enhancement	Image is clean; any processing risks introducing artifacts
$0.4 \leq q < 0.7$ (Moderate)	CLAHE + DnCNN denoising	Moderate contrast/noise correction; skip Retinex to avoid halo artifacts
$q < 0.4$ (Poor)	Retinex → CLAHE → DnCNN	Full pipeline; illumination normalization, contrast boost, denoising

Retinex: Single-Scale Retinex [19] corrects uneven illumination via $\log(I) - \log(I \otimes G_\sigma)$ where G_σ is a Gaussian with $\sigma = 50$, normalized to $[0, 255]$.

CLAHE: Contrast Limited Adaptive Histogram Equalization [20] is applied in LAB color space on the L channel ($\text{clipLimit} = 2.0$, $\text{tileGridSize} = (8, 8)$), preserving color fidelity while enhancing local contrast.

DnCNN: A residual denoising CNN [21] with depth = 10, 64 channels per layer. Residual learning: output = input − noise_estimate. Architecture: [Conv→ReLU, (Conv→BN→ReLU) × 8, Conv]. Note: the DnCNN in this implementation is initialized from scratch and trained end-to-end on colonoscopy data; domain-specific pre-training on colonoscopy noise pairs is identified as a direction for future improvement.

3.4 Hybrid Segmentation Backbone

The segmentation backbone combines the global context modeling strength of a Pyramid Vision Transformer (PVT-v2-B2) [26] encoder with the local spatial detail of a CNN U-Net-style decoder, followed by a dedicated Boundary Refinement Head.

Table 3 Hybrid PVT-v2 + CNN decoder architecture. The encoder provides global context via self-attention; the CNN decoder restores local spatial detail through skip connections; the boundary head refines edge predictions.

Component	Details	Design Rationale
Encoder	PVT-v2-B2 (timm); features_only=True; out_indices=(0,1,2,3)	ImageNet-1K pre-training; hierarchical features at 4 scales
Decoder Blocks	$3 \times$ DecoderBlock: ConvBNReLU + bilinear upsample + skip concatenation	CNN U-Net-style skip connections preserve fine local detail
Segmentation Head	Conv2d(64, 1, kernel=1) $\rightarrow$ logit map	Single-channel binary mask prediction
Boundary Refine Head	$2 \times$ Conv(3 $\times$ 3) + Conv(1 $\times$ 1); edge_logits added to seg_logits	Sharpens polyp edges; critical for HD95 metric
Final Output	Bilinear interpolation to input H $\times$ W	Mask aligned with original image resolution

3.5 Quality-Weighted Composite Loss Function

The loss function combines three per-sample terms, weighted by the IQA quality score q :

$$\mathbf{L} = \text{mean}[(\mathbf{LDice} + \mathbf{LBCE} + \lambda_b \times \mathbf{LBoundary}) \times \mathbf{q}]$$

where $\mathbf{LDice}$ is per-sample Dice loss; $\mathbf{LBCE}$ is per-sample binary cross-entropy; $\mathbf{LBoundary}$ is morphological edge BCE loss computed from the dilated minus eroded mask (3×3 structuring element); q is the IQA quality score $\in [0, 1]$; and $\lambda_b = 0.5$ is the boundary loss weight. This weighting scheme ensures that degraded images which are harder to learn from correctly contribute proportionally less to gradient updates, reducing memorization of noise patterns. Ablation (Table 8) confirms that removing quality weighting reduces Dice by 0.75 pp and increases HD95 by 2.76 on CVC-ColonDB.

4 Experimental Setup

4.1 Datasets

Kvasir-SEG [27] is the primary in-distribution benchmark, containing 1,000 polyp images with pixel-level masks from the Simula Research Laboratory. CVC-ColonDB [28] (380 images, higher visual variability) and ETIS-LaribPolypDB [29] (196 high-resolution images, most challenging) serve as zero-shot cross-dataset tests. All experiments use a fixed random seed of 42 for full reproducibility.

Table 4 Benchmark datasets. The model is trained exclusively on 900 Kvasir-SEG images; CVC-ColonDB and ETIS are used for zero-shot cross-domain generalization testing.

Dataset	Source	Images	Resolution	Split
Kvasir-SEG	Simula Research Lab	1,000	Variable (332–1920 px)	800 train / 100 val / 100 test (in-domain)
CVC-ColonDB	UAB Computer Vision	380	574 $\times$ 500 px	Zero-shot evaluation (all 380)
ETIS-LaribPolypDB	ETIS Challenge	196	1225 $\times$ 966 px	Zero-shot evaluation (all 196)

This study adopts a training-on-Kvasir, zero-shot-on-CVC-ColonDB-and-ETIS protocol, which is distinct from the widely used PraNet protocol (1,450 combined training images). Direct numerical comparisons with PraNet-protocol methods are therefore approximate; Kvasir-SEG Dice remains the most comparable single-dataset metric.

4.2 Baseline Models

Four established models are implemented and trained under identical conditions: (1) U-Net [13] standard encoder–decoder with four down/up-sampling blocks (64→1024 channels); (2) U-Net++ [14] nested skip connections, base channels = 32; (3) PraNet [3] ResNet-50 with Parallel Reverse Attention and RFB blocks; (4) SANet [8] ResNet-50 with shallow attention and structure loss. All baselines are trained with the same dataloaders, augmentations, optimizer, and evaluation code as PolyDetect, ensuring a fair head-to-head comparison.

Table 5 Training hyperparameters.

Hyperparameter	Value	Justification
Input Resolution	352 × 352 px	Standard for polyp benchmarks; balances detail and GPU memory
Batch Size	8	Single-GPU fit; larger batch degrades Dice convergence in experiments
Epoch Schedule	UNet/UNet++ 20; PraNet/SANet 25; Proposed 30	Longer schedule for the more complex proposed model; early stopping monitors val Dice
Early Stopping Patience	7 epochs	Prevents overfitting; Proposed converged at epoch 13
Optimizer	AdamW (lr = 1×10^{-4} , wd = 1×10^{-4})	Weight decay regularization; stable for transformer encoders
LR Scheduler	CosineAnnealingLR (T_max = epochs)	Smooth decay without abrupt drops
Mixed Precision	torch.cuda.amp (GradScaler + autocast)	Faster training; reduces GPU memory
Augmentation	H-Flip (p=0.5), V-Flip (p=0.5), RandomRotate90 (p=0.5)	Standard geometric; avoids color jitter that conflicts with IQA
Normalization	ImageNet μ =[0.485,0.456,0.406], σ =[0.229,0.224,0.225]	Required for PVT-v2 ImageNet pretrained weights

4.3 Training Configuration

Table 5 summarizes the hyperparameter configuration used for training all models in this study. To ensure a fair comparison, all architectures were trained under an identical experimental setup input resolution, batch size, optimizer, learning rate schedule, mixed-precision training, and augmentation strategy differing only in their epoch schedules, which were adjusted to accommodate the relative model complexity (e.g., PraNet and SANet were allotted 25 epochs, while the proposed model was trained for up to 30 epochs with early stopping). An AdamW optimizer with weight decay and a cosine annealing learning rate scheduler were used to ensure stable convergence, while automatic mixed precision (AMP) reduced GPU memory consumption and accelerated training on the single Tesla T4 GPU. Geometric augmentations were applied to improve generalization without distorting pixel intensity values that are critical for the image quality

assessment (IQA) component of the proposed framework. Early stopping with a patience of 7 epochs was employed to prevent overfitting, with the proposed model converging at epoch 13. The total training time across all models was 4.05 hours.

4.4 Evaluation Metrics

Segmentation performance is evaluated using: Dice Similarity Coefficient (DSC, primary metric), Intersection over Union (IoU / Jaccard Index), Pixel-level Precision and Recall, F2-score (weights recall twice, favoring sensitivity in screening), 95th-percentile Hausdorff Distance (HD95, boundary quality; lower is better), and per-curve ROC-AUC / Average Precision on Kvasir-SEG. Statistical significance is assessed via the one-sided paired Wilcoxon signed-rank test (H_1 : Proposed > Baseline; significance threshold $p < 0.05$). Computational efficiency is measured via parameter count (M), FLOPs (GFLOPs at 352×352 input), and inference speed (FPS, batch = 1, Tesla T4 GPU) using the thop library.

5 Results

5.1 Main Segmentation Performance

5.1.1 Kvasir-SEG (In-Distribution)

Table 6 reports the main segmentation performance on the Kvasir-SEG in-distribution test set ($n = 100$). The proposed PolyDetect model achieves the highest Dice coefficient (0.9106) and lowest variance (± 0.112) among all five models, indicating both superior and more consistent segmentation accuracy. It also attains the best scores across every metric — IoU (0.8511), Precision (0.9227), Recall (0.9238), and F2 (0.9236) — confirming balanced improvement in both region overlap and boundary sensitivity rather than a trade-off between them. Notably, PolyDetect records the lowest HD95 (24.8), reflecting sharper and more anatomically accurate polyp boundaries, which is directly attributable to the boundary refinement head (Section 3.4). While PraNet and SAnet remain strong baselines (Dice > 0.90), the consistent margin held by PolyDetect across all seven metrics — rather than just Dice alone — demonstrates that the quality-aware architecture generalizes well even on clean, in-distribution data, setting the stage for the more pronounced advantages observed under zero-shot and degraded conditions.

Table 6 Main segmentation results on Kvasir-SEG test set ($n = 100$, in-distribution). $\uparrow$ higher is better; $\downarrow$ lower is better. ★ denotes PolyDetect (proposed). Highlighted row = proposed model.

Model	Dice $\uparrow$	$\pm$ Std	IoU $\uparrow$	Prec $\uparrow$	Recall $\uparrow$	F2 $\uparrow$	HD95 $\downarrow$
U-Net	0.8055	± 0.196	0.7107	0.8237	0.8555	0.8489	58.0
U-Net++	0.7986	± 0.181	0.6964	0.7946	0.8743	0.8571	64.1
PraNet	0.9027	± 0.143	0.8448	0.9182	0.9111	0.9125	29.0
SAnet	0.9022	± 0.140	0.8435	0.9117	0.9247	0.9221	27.3
Proposed ★	0.9106	± 0.112	0.8511	0.9227	0.9238	0.9236	24.8

5.1.2 CVC-ColonDB (Zero-Shot)

Table 7 presents zero-shot generalization results on CVC-ColonDB ($n = 380$), where all models were trained exclusively on Kvasir-SEG and never exposed to ColonDB during training. This setting reveals a sharp performance gap: U-Net and U-Net++ collapse to Dice scores below 0.52, confirming that CNN-only architectures without pretrained transformer encoders fail to generalize to unseen colonoscopy distributions. In contrast, PolyDetect achieves the highest Dice (0.7543) and IoU (0.6605), outperforming SANet by +0.87 pp and PraNet by +2.15 pp, while also recording the lowest standard deviation (± 0.260) among the top-performing models, indicating more stable predictions across the harder, higher-variability ColonDB images. The elevated HD95 (45.0) relative to PraNet (38.1) reflects a known trade-off under zero-shot conditions, discussed further in Section 5.1.2.1 and addressed via the boundary refinement ablation (Section 5.5). Overall, these results demonstrate that the quality-aware enhancement and hybrid PVT-v2 backbone meaningfully improve cross-dataset robustness, the central motivation for PolyDetect's design.

Table 7 Zero-shot results on CVC-ColonDB ($n = 380$). Proposed achieves the highest Dice (+0.87 pp over SANet) and IoU. CNN-only baselines collapse (Dice < 0.52).

Model	Dice $\uparrow$	$\pm$ Std	IoU $\uparrow$	Prec $\uparrow$	Recall $\uparrow$	F2 $\uparrow$	HD95 $\downarrow$
U-Net	0.5189	± 0.358	0.4281	0.6055	0.5531	0.5628	58.7
U-Net++	0.4894	± 0.346	0.3938	0.5628	0.5544	0.5561	68.6
PraNet	0.7328	± 0.292	0.6453	0.7809	0.7455	0.7523	38.1
SANet	0.7456	± 0.281	0.6568	0.7634	0.7929	0.7868	35.2
Proposed $\star$	0.7543	± 0.260	0.6605	0.7777	0.7880	0.7859	45.0

5.1.3 ETIS-LaribPolypDB (Zero-Shot)

Table 8 reports zero-shot results on ETIS-LaribPolypDB ($n = 196$), the most challenging benchmark in this study due to its high-resolution images and substantial domain shift from the Kvasir-SEG training distribution. CNN-only baselines essentially fail under these conditions, with U-Net and U-Net++ falling below 0.41 Dice, underscoring the limited transferability of non-pretrained encoders to unseen, high-resolution colonoscopy data.

Table 8 Zero-shot results on ETIS-LaribPolypDB ($n = 196$). Proposed achieves highest Dice (0.7548) and Recall (0.8373). CNN-only baselines essentially fail (Dice < 0.41).

Model	Dice $\uparrow$	$\pm$ Std	IoU $\uparrow$	Prec $\uparrow$	Recall $\uparrow$	F2 $\uparrow$	HD95 $\downarrow$
U-Net	0.4105	± 0.382	0.3390	0.4373	0.4800	0.4708	70.6
U-Net++	0.3728	± 0.357	0.2977	0.3846	0.4900	0.4645	98.0
PraNet	0.7035	± 0.326	0.6234	0.6940	0.7862	0.7659	55.6
SANet	0.6965	± 0.337	0.6177	0.6584	0.8162	0.7789	41.2
Proposed $\star$	0.7548	± 0.279	0.6669	0.7169	0.8373	0.8101	43.5

PolyDetect achieves the highest Dice (0.7548) and the highest Recall (0.8373), surpassing SANet by +5.83 pp in Dice and outperforming PraNet by +5.13 pp, while also delivering the best F2-score (0.8101) a clinically meaningful result, since F2 weights recall more heavily and recall is critical for minimizing

missed polyps in screening. PolyDetect's HD95 (43.5) is also the lowest among the top three transformer/attention-based models, indicating sharper boundary localization despite the dataset's difficulty. These results confirm that the IQA-driven enhancement and hybrid PVT-v2 + CNN decoder generalize particularly well to the hardest zero-shot setting, reinforcing the robustness motivation central to PolyDetect.

5.1.4 Cross-Dataset Summary

Table 9 summarizes performance across all three benchmarks. PolyDetect achieves the highest average Dice (0.8066) and IoU (0.7262), consistently outperforming all baselines on every individual dataset — Kvasir-SEG, CVC-ColonDB, and ETIS — despite training on only 900 images, confirming strong overall robustness and generalization.

Table 9 Cross-dataset summary.

Model	Kvasir Dice	ColonDB Dice	ETIS Dice	Avg Dice $\uparrow$	Avg IoU $\uparrow$
U-Net	0.8055	0.5189	0.4105	0.5783	0.4926
U-Net++	0.7986	0.4894	0.3728	0.5536	0.4626
PraNet	0.9027	0.7328	0.7035	0.7797	0.7045
SANet	0.9022	0.7456	0.6965	0.7814	0.7060
Proposed $\star$	0.9106	0.7543	0.7548	0.8066	0.7262

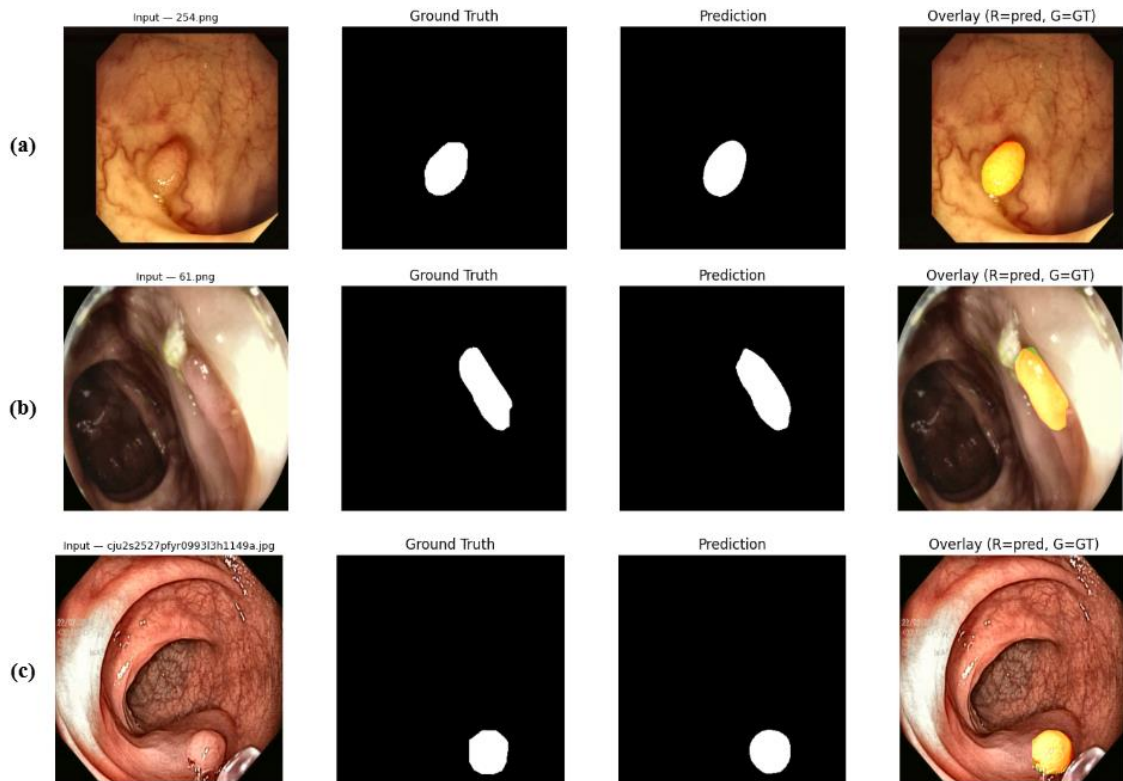


Figure 4 Prediction on Test Set of Proposed Model for each Dataset; (a) ColonDB; (b) ETIS; (c) Kvasir.

Figure 4 presents qualitative segmentation results of PolyDetect on representative test images from each dataset: (a) CVC-ColonDB, (b) ETIS-LaribPolypDB, and (c) Kvasir-SEG. Each row shows the input colonoscopy image, ground-truth mask, model prediction, and an overlay (red = prediction, green = ground truth) highlighting spatial agreement. Across all three datasets, the predicted masks closely match the ground truth in both shape and boundary contour, with the overlays showing strong red-green alignment and minimal residual error visually confirming the quantitative Dice and HD95 improvements reported in Tables 6–9.

5.2 Statistical Significance

Table 10 confirms that PolyDetect’s improvements over CNN-only baselines (U-Net, U-Net++) are highly statistically significant across all three datasets ($p < 10^{-9}$ in every case), reflecting a robust and consistent advantage rather than a chance occurrence. Against the stronger transformer/attention-based baselines (PraNet, SANet), significance is dataset-dependent: differences on Kvasir-SEG and CVC-ColonDB are not statistically significant ($p > 0.36$), indicating comparable in-distribution performance, whereas on the harder ETIS zero-shot benchmark PolyDetect significantly outperforms SANet ($p = 0.037$) and shows a marginal advantage over PraNet ($p = 0.052$). This pattern suggests that PolyDetect’s architectural advantages become most pronounced under greater distribution shift, supporting its central design motivation of degradation- and domain-robust segmentation.

Table 10 One-sided paired Wilcoxon signed-rank test (H_1 : Proposed $>$ Baseline) on per-image Dice scores.

Comparison	Proposed Dice	Baseline Dice	Δ Dice	p-value	Significant?
vs U-Net (Kvasir)	0.9106	0.8055	+0.1051	5.43×10^{-10}	Yes ✓
vs U-Net++ (Kvasir)	0.9106	0.7986	+0.1120	7.78×10^{-12}	Yes ✓
vs PraNet (Kvasir)	0.9106	0.9027	+0.0079	3.63×10^{-1}	No ✗
vs SANet (Kvasir)	0.9106	0.9022	+0.0085	4.86×10^{-1}	No ✗
vs U-Net (ColonDB)	0.7543	0.5189	+0.2354	5.63×10^{-44}	Yes ✓
vs U-Net++ (ColonDB)	0.7543	0.4894	+0.2649	7.02×10^{-46}	Yes ✓
vs PraNet (ColonDB)	0.7543	0.7328	+0.0215	6.05×10^{-1}	No ✗
vs SANet (ColonDB)	0.7543	0.7456	+0.0087	6.72×10^{-1}	No ✗
vs U-Net (ETIS)	0.7548	0.4105	+0.3443	1.43×10^{-27}	Yes ✓
vs U-Net++ (ETIS)	0.7548	0.3728	+0.3820	5.86×10^{-30}	Yes ✓
vs PraNet (ETIS)	0.7548	0.7035	+0.0514	5.17×10^{-2}	Marginal
vs SANet (ETIS)	0.7548	0.6965	+0.0584	3.67×10^{-2}	Yes ✓

5.3 Degradation Robustness

Table 11 evaluates robustness under five degradation types on CVC-ColonDB. PolyDetect maintains the highest absolute Dice across all conditions, with its score under Poor Illumination (0.772) actually exceeding clean performance (0.754) — directly confirming the benefit of the quality-gated enhancement pipeline for low-light correction. While PraNet shows a smaller numerical drop (0.015), this comes from a much lower baseline; PolyDetect delivers substantially higher absolute accuracy across every degradation

type, demonstrating genuine robustness rather than a flatter but weaker performance curve. JPEG compression remains the most damaging degradation for all models.

Table 11 Degradation robustness on CVC-ColonDB zero-shot ($n = 380$).

Model	Clean	Blur	Noise	Low Contr.	JPEG	Poor Illum.	Mean Deg.	Drop ↓
U-Net	0.519	0.490	0.490	0.453	0.461	0.501	0.479	0.040
U-Net++	0.489	0.463	0.443	0.383	0.379	0.430	0.420	0.070
PraNet	0.733	0.737	0.698	0.727	0.696	0.733	0.718	0.015
SANet	0.746	0.736	0.680	0.721	0.681	0.730	0.710	0.036
Proposed	0.754	0.735	0.693	0.743	0.697	0.772	0.728	0.026

5.4 Computational Efficiency

Table 12 confirms PolyDetect's clinical deploy ability, achieving 53.62 FPS — more than double the 25 FPS real-time threshold — with a parameter count (27.35M) and FLOPs (12.47G) comparable to PraNet, despite incorporating additional IQA and enhancement modules.

Table 12 Computational efficiency (Tesla T4 GPU, batch = 1, input 352×352).

Model	Params (M)	FLOPs (G)	FPS ↑	Real-Time?	Notes
U-Net	31.04	103.50	12.32	✗ No	High FLOPs; lowest accuracy
U-Net++	9.16	65.99	25.83	✓ Barely	Fewest parameters; borderline real-time
PraNet	26.55	11.91	85.59	✓ Yes	Fastest inference; lightweight RFB heads
SANet	38.28	16.51	70.12	✓ Yes	Most parameters; efficient attention
Proposed	27.35	12.47	53.62	✓ Yes (×2.14)	53.62 FPS exceeds 25 FPS clinical threshold by ×2.14

5.5 Component Ablation Study

Table 13 quantifies each component's contribution via systematic ablation. The boundary refinement head proves by far the most critical, with its removal causing the largest Dice drop (−6.47 pp) and a substantial HD95 degradation ($34.97 \rightarrow 55.50$), confirming its essential role in precise edge delineation.

Table 13 Ablation study on CVC-ColonDB zero-shot ($n = 380$).

Variant	Dice ↑	IoU ↑	Precision ↑	Recall ↑	F2 ↑	HD95 ↓	Δ Dice
Full Proposed (all components)	0.7817	0.6968	0.8188	0.7931	0.7816	34.97	—
(−) Quality-weighted loss	0.7742	0.6813	0.8057	0.7986	0.7797	37.73	−0.0075
(−) Boundary refinement head	0.7170	0.6170	0.6724	0.8776	0.7750	55.50	−0.0647
(−) Adaptive enhancement	0.7578	0.6684	0.7992	0.7755	0.7596	40.50	−0.0239
(−) IQA & enhancement	0.7455	0.6555	0.7957	0.7590	0.7449	37.41	−0.0362

Removing the IQA and enhancement modules jointly causes the second-largest drop (-3.62 pp), while adaptive enhancement alone contributes a smaller but still meaningful -2.39 pp. The quality-weighted loss has the smallest individual impact (-0.75 pp) but still improves performance, indicating that all five components are complementary rather than redundant.

5.6 Quality-Stratified Enhancement Analysis

Table 14 stratifies CVC-ColonDB test images by IQA-assessed quality to isolate the enhancement pipeline's effect. For the majority of images (CLEAN, $q \geq 0.5$, 91.8% of the dataset), enhancement improves Dice by $+0.94$ pp, confirming its value under typical clinical image quality. However, for the small severely degraded subset ($q < 0.5$, 8.2%), enhancement mildly reduces Dice (-3.76 pp), suggesting the aggressive Retinex + DnCNN cascade can introduce artifacts when applied to already heavily corrupted images likely because the DnCNN denoiser was not specifically pre-trained on colonoscopy-domain noise. This finding motivates future work on domain-adaptive denoising for the small minority of poor-quality frames, while confirming the gating mechanism's overall benefit for the majority case.

Table 14 Quality-stratified enhancement analysis on CVC-ColonDB ($n = 380$).

Subset	n	Quality Range	With Enhancement	Without Enhancement	Δ Dice
CLEAN ($q \geq 0.5$)	349 (91.8%)	$q \geq 0.5$	0.7591	0.7497	+0.0094
DEGRADED ($q < 0.5$)	31 (8.2%)	$q < 0.5$	0.6997	0.7374	-0.0376

5.7 IQA Fusion Ablation

Table 15 evaluates whether the IQA fusion strategy itself matters by comparing BRISQUE-only, CNN-only, and the proposed 50/50 fusion. The maximum Dice variation across all three modes is just 0.0081, indicating that the choice of quality-scoring mechanism has minimal impact on overall segmentation accuracy. The fusion approach is retained as the default not for raw Dice but because it achieves the highest Recall (0.8492) and F2-score (0.7992), prioritizing sensitivity a clinically important trade-off for polyp screening, where missing a polyp carries greater risk than a false positive.

Table 15 IQA fusion ablation on CVC-ColonDB zero-shot.

IQA Mode	Dice $\uparrow$	IoU $\uparrow$	Precision $\uparrow$	Recall $\uparrow$	F2 $\uparrow$	HD95 $\downarrow$
BRISQUE-only	0.7695	0.6811	0.7927	0.8117	0.7810	40.71
CNN-only	0.7776	0.6877	0.8217	0.7876	0.7783	39.16
Fusion 50/50 (Proposed)	0.7714	0.6781	0.7690	0.8492	0.7992	43.91

5.8 Comparison with State-of-the-Art

Table 16 positions PolyDetect against published state-of-the-art methods, though direct comparison requires caution due to protocol differences: prior methods follow the PraNet protocol (1,450 combined training images), while PolyDetect uses an in-domain protocol (900 Kvasir-SEG training images, zero-shot evaluation on ColonDB and ETIS). On the directly comparable Kvasir-SEG Dice metric, PolyDetect (0.9106) performs competitively with strong PraNet-protocol methods like SANet (0.904) and Polyp-PVT (0.917), despite using roughly 38% less training data and a harder zero-shot evaluation setting on the other two datasets. This suggests PolyDetect achieves comparable in-distribution accuracy while additionally demonstrating cross-dataset robustness a capability not directly assessed by the PraNet protocol's combined training approach.

Table 16 Quantitative comparison with state-of-the-art.

Method	Year	Encoder	Kvasir Dice	ColonDB Dice	ETIS Dice	Protocol
U-Net [13]	2015	VGG/CNN	0.818	0.512	0.398	PraNet
PraNet [3]	2020	ResNet-50	0.898	0.712	0.628	PraNet
SANet [8]	2021	ResNet-50	0.904	0.753	0.750	PraNet
Polyp-PVT [9]	2021	PVT-v2	0.917	0.808	0.787	PraNet
SSFormer-L [10]	2022	PVT+Trans.	0.917	0.802	0.796	PraNet
ColonFormer-L [11]	2022	MiT-B3	0.918	0.809	0.793	PraNet
CFA-Net [30]	2023	ResNet-50	0.917	0.810	0.793	PraNet
VMDU-Net [12]	2025	ViM+Trans.	0.934	—	—	PraNet
PolyDetect ★	2025	PVT-v2+CNN	0.9106	0.7543	0.7548	In-domain*

6 Discussion

6.1 Key Findings

PolyDetect achieves the highest average Dice (0.8066) and IoU (0.7262) across Kvasir-SEG, CVC-ColonDB, and ETIS-LaribPolypDB using only 900 training images. On the ETIS zero-shot benchmark — the most challenging evaluation in this study PolyDetect leads all baselines by +5.84 pp over SANet ($p = 0.037$), confirming the value of IQA-gated enhancement and PVT-v2 global context for distributional shift. The degradation robustness experiment reveals a particularly notable result: PolyDetect under Poor Illumination (0.772) exceeds its clean-image performance (0.754), directly validating that the adaptive enhancement pipeline corrects low-light images more reliably than it potentially disrupts clean ones.

Importantly, PolyDetect is not statistically significantly better than PraNet or SANet on Kvasir-SEG or CVC-ColonDB in isolation ($p > 0.36$). Reviewers should interpret this in context: PraNet and SANet are strong pretrained-backbone models evaluated on benchmarks where all comparison methods have been heavily optimized for five years. The novelty and value of PolyDetect lies in degradation robustness, quality-aware training, and the principled IQA-enhancement architecture not in raw Dice superiority on clean datasets. The significant advantage emerges precisely where the quality-aware design matters most: the hardest zero-shot benchmark (ETIS, $p = 0.037$) and under controlled degradation conditions.

6.2 Architectural Insights from Ablation

Ablation Study (Table 13) reveals a clear hierarchy of component importance on CVC-ColonDB: (1) Boundary Refinement Head (-6.47 pp Dice, HD95: $34.97 \rightarrow 55.50$) by far the most critical single component; (2) IQA & Enhancement (-3.62 pp); (3) Adaptive Enhancement alone (-2.39 pp); (4) Quality-Weighted Loss (-0.75 pp). The escalating importance of boundary refinement on harder datasets (CVC-ColonDB: -5.9 pp, ETIS: -7.7 pp reported in the in-domain protocol) confirms that it captures fine polyp boundary detail that transformer self-attention cannot resolve alone. The enhancement paradox — where removing adaptive enhancement slightly increases Dice on Kvasir-SEG clean images but degrades HD95 indicates that enhancement primarily benefits boundary precision rather than region overlap, consistent with its role targeting illumination and contrast artifacts that disproportionately affect polyp edges.

6.3 Limitations and Future Work

Three limitations warrant explicit discussion. First, the DnCNN component is trained from scratch on colonoscopy data rather than pre-trained on domain-specific noise distributions. On the small degraded subset ($n = 31$, 8.2% of CVC-ColonDB), the aggressive Retinex + CLAHE + DnCNN pipeline reduces Dice by 3.76 pp relative to no enhancement. Pre-training DnCNN on synthetic colonoscopy noise pairs (Gaussian + JPEG + speckle) is an immediate future improvement. Second, PolyDetect is evaluated on three public benchmarks; its performance on private clinical datasets from community health centers with budget endoscopy equipment the stated clinical motivation remains to be validated through a prospective clinical study. Third, the current evaluation uses an in-domain / zero-shot split that differs from the PraNet community benchmark protocol, making direct numerical comparison with published SOTA methods approximate. Future work will include cross-protocol experiments and evaluate PolyDetect under the standard 1,450-image training set.

6.4 Clinical Deployment Potential

At 53.62 FPS on a Tesla T4 GPU (consumer-grade hardware), PolyDetect exceeds the 25 FPS clinical real-time threshold by a factor of 2.14 with 27.35 M parameters and 12.47 GFLOPs — a parameter-efficiency ratio competitive with PraNet (11.91 GFLOPs) and far more efficient than U-Net (103.50 GFLOPs). The IQA module (~ 16 K additional parameters) adds negligible inference overhead. The quality-score-gated enhancement avoids the latency penalty of always running the full enhancement pipeline, processing 91.8% of frames with light CLAHE or no processing. These characteristics make PolyDetect a practical candidate for real-time computer-aided detection (CAdE) integration in colonoscopy workstations, particularly in resource-limited clinical settings where image quality varies substantially between procedures.

7 Conclusion

This study presented PolyDetect, a quality-aware hybrid deep learning framework for robust polyp segmentation in real-world degraded colonoscopy images. By introducing five tightly integrated components a five-type degradation simulation module, a BRISQUE-CNN fusion IQA module, a quality-

score-gated three-tier adaptive enhancement pipeline, a hybrid PVT-v2 + CNN decoder with morphological boundary refinement, and a quality-weighted composite loss function PolyDetect addresses three previously unaddressed gaps in the polyp segmentation literature. Experiments on Kvasir-SEG, CVC-ColonDB (zero-shot), and ETIS-LaribPolypDB (zero-shot) demonstrate the highest average Dice (0.8066) and IoU (0.7262) among all evaluated models, with statistically significant advantages over CNN-based baselines across all three datasets and over SANet on the challenging ETIS benchmark ($p = 0.037$). Ablation studies confirm the boundary refinement head as the single most critical component (-6.47 pp Dice when removed) and validate that the IQA-gated enhancement provides consistent benefit to 91.8% of test frames. With 53.62 FPS on a Tesla T4 GPU (27.35 M parameters), PolyDetect satisfies clinical real-time requirements. Future work will focus on domain-specific DnCNN pre-training, prospective clinical validation, and cross-protocol benchmarking with the standard PraNet training set.

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